

Instructor: Date:

Learner: Regional Branch:

Program: Center:

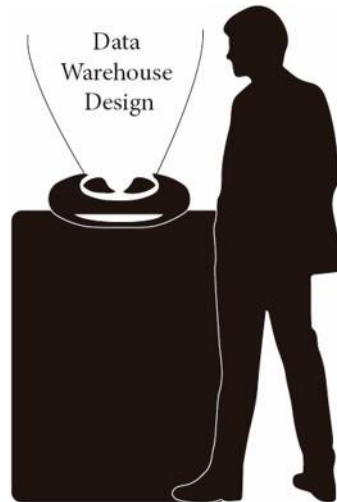
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Let's Start!

Elements of Data Warehouse



Learning Outcome:

Designs the data structure based on the database structures and its characteristics.

Understand the concept of data warehouse, its components and characteristics.

Can Do Statements:

- I can understand written information about what the data warehouse contained.
- I can answer questions related with the components of a data warehouse.
- I can list important information about the design behind a data warehouse.

Ask your partner:

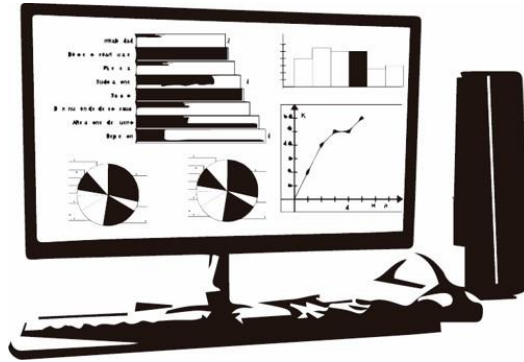
- What do you think are the most basic elements in a data warehouse? Share your thoughts.
- What are some important aspects when designing a warehouse?





Let's Explore!

What are the components of a Data Warehouse? What about its design?



Instructions: Read the text below.

DATA WAREHOUSE COMPONENTS AND DESIGN YOU MIGHT WANT TO READ ABOUT

There were some important features I didn't take into account when I designed a data warehouse. I needed to include aspects like data collections, tables, table help, data elements, data element help box, and summary data. You'd better check the brief descriptions, which I've made, about the features previously mentioned for you to keep them in mind when design time comes.

Data collections contain information regarding transaction systems. Tables include elements which describe items depending of their importance which helps out when finding facts. Table help gives information about common uses and related tables, and data elements have items that cannot be divided.

Maybe, you're full of questions about components of data warehouse. Based on my experience, I've created and answered a couple of questions that might be of great importance to you: What are the components of data warehouse? There are seven main components you ought to include when creating data warehouse: database, transformation tools, Meta data, access tools, data Marts, delivery system, and administration & management. Why have the transformation tools been of great relevance for data warehouse implementation? Well, they've been of great value because by sourcing, acquiring and cleaning up data from operational systems, data is organized in an appropriate format to be uploaded to the apps that run off the warehouse. You'd better define what the data dimensions and measures will be in order to succeed with its implementation.

Finally, you might want to take a look at some design tips: any design method is inoperative if tool support hasn't been done. You ought to perform a conceptual design, bound workload, figure out data magnitude, implement a logical design, and come up with documentation. Remember, always ask yourself: Did I follow the design tips?





Let's Practice!

Take a look at some important information about the design behind a data warehouse.

Instructions: Write the words to correctly complete the sentences. Use words from the box below.

might want to / want to figure out / You had
You ought / had better implement

1. You _____ take a look at some design tips.
2. _____ better consider that any design method is inoperative if the tool support hasn't been done.
3. _____ to perform a conceptual design.
4. You might _____ the data magnitude.
5. You _____ a logical design and come up with important documentation.



Let's Practice!

Let's identify important information about the components of a data warehouse.

Instructions: Match the sentences from Column A with Column B.

COLUMN A

1. How many components have been included in the data warehouse?
2. Why have the transformation tools been of great help?
3. How has data warehouse been organized?
4. Why have the data dimensions been defined?
5. Which has been the purpose of defining the data measures?

COLUMN B

- a. Data dimensions have been defined in order to succeed with its implementation.
- b. The data dimensions and measures have been defined to facilitate their implementation.
- c. The components have helped to acquire and clean up the data from operational systems.
- d. The data has been organized in an appropriate format to be uploaded to the apps.
- e. There are seven components that have been included in the data warehouse.



Extension Activity

Now it's your turn to design a data warehouse.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Find information about how to design a data warehouse.
- Choose the most important information.
- Organize the information in a flowchart.
- Explain to your partners the process of designing a data warehouse.

Self-Assessment

Yes

No

- I find information about how to design a data warehouse.
- I choose the most important information about designing a data warehouse.
- I organize the information in a flowchart.
- I explain to my partners the process of designing a data warehouse.

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Target Words

- **run off** (phrasal verb): to execute
e.g. Well, they've been of great value because by sourcing, acquiring and cleaning up data from operational systems, data is organized in an appropriate format to be uploaded to the apps that run off the warehouse.
- **delivery** (noun): a shipment of goods
e.g. There are seven main components you ought to include when creating data warehouse: database, transformation tools, Meta data, access tools, data Marts, delivery system, and administration & management.
- **relevance** (noun): something that is pertinent
e.g. Why have the transformation tools been of great relevance for data warehouse implementation?
- **inoperative** (adjective): that is useless
e.g. Finally, you might want to take a look to some design tips: any design method is inoperative if tool support hasn't been done.
- **summary** (noun): a brief but complete abstract of something
e.g. I needed to include aspects like data collections, tables, table help, data elements, data element help box, and summary data.
- **fact** (noun): something that exists or happens
e.g. Tables include elements which describe items depending of their importance which helps out when finding facts.



Learning Points

Present Perfect vs Simple Past

You can use the **present perfect** to refer to past events that are not finished yet. On the other hand, we use the **simple past** to refer to finished actions in the past.

To create a statement using the **present perfect**, you need to use the auxiliary verbs **HAVE** and **HAS**, and the action verb in its **past participle form**. Check the following structure:

Subject + **auxiliary verb (has/have)** + **action verb (past participle form)** + complement

Example:

*I **have designed** a well-structured warehouse.*

To create a negative statement using the present perfect, you need to use the auxiliaries HAVE (I-you-we-they) and HAS (he-she-it) and the particle NOT with an action verb in its past participle form.

The structure we use is:

Subject + **auxiliary verb (has/have)** + **not** + **action verb (past participle form)** + complement.

Example:

*We **haven't included** all of the warehouse features.*

To create a question, you need to change the position of the auxiliary verb with the one of the subject. Plus, you need to add a question mark. Now, remember that we make that change when we want to make a yes/no question; plus, take into account that we can add a wh- word at the beginning to make an open question (whose answer is NOT yes or no). Check the structure that we use:

(Wh- question) + Auxiliary verb (has/have) + subject + action verb (past participle form) + complement + ?

Example:

***Have** you **read** all of the data warehouse components?(yes/no question)*

***What have** you **done** to learn about the data warehouse components? (open question)*

To create a statement using the **simple past**, you need to use the verb in its past form; remember there are two types of verbs REGULAR (verbs that you add -ED at the end) and IRREGULAR (verbs that change in spelling and pronunciation).





Learning Points

The structure that we use is the following:

Subject + **verb (past form)** + complement.

Example:

*I **used** all of the features for the warehouse design. (regular verb)*

*I **began** to work on this data warehouse last year. (irregular verb)*

To create a negative statement using **simple past**, you need to use the auxiliary verb **did** and the negative particle **NOT** with a verb in its **base form**.

The structure that we follow is :

Subject + auxiliary verb (did) + not + action verb in base form + complement.

Example:

*I **did not know** what is behind data warehouse design.*

*I **didn't understand** what data warehouse was about.*

To make a yes /no question, you need to start the question with the auxiliary verb did. On the other hand, if you want to make an open question, you have to use a wh- word plus the auxiliary verb. Check the following structure:

(Wh – word) + did + subject + verb in its base form + complement + ?

Example:

***Did** you **check** all of the transformation tools?*

***Where did** you **learn** about all?*

Examples

- *You'd better check the brief descriptions, which I've made, about the features previously mentioned for you to keep them in mind when design time comes.*
- *Finally, you might want to take a look at some design tips: any design method is inoperative if tool support hasn't been done.*
- *There were some important features I didn't take into account when I designed a data warehouse.*
- *Remember, always ask yourself: Did I follow the design tips?*



Learning Points

Data Warehouse Components

The complete architecture for **data warehouse** is made up of some important components that need to be included when designing.

A data warehouse is built under seven different main components: database, transformation tools, Meta Data, access tools, Data Marts, information delivery system, and data warehouse administration & management.

Database is the key feature for a warehouse since databases are always used in the management systems. Transformation tools are of great value because by sourcing, acquiring and cleaning up data from operational systems can be put on an appropriate format to be uploaded to the apps that run off the warehouse.

Meta Data refers to data that draws the data warehouse. It is mainly used for building, maintaining and using a data warehouse. It is classified into technical meta data and business meta data. Access tools are all of the possible ways that users have for interacting with the data warehouse.

Data Marts are those different partitions that the warehouse can get. The most relevant information in the Data Marts is obtained from the warehouse. Regarding the information delivery system, we can say that it basically moves the required information to different destinations whenever it is scheduled.

Finally, data warehouse administration & management relates to its security and the management of information.